Homework 10

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Problem 1

Prove Theorem 4.24: self-adjoint, skew-self-adjoint and unitary transformations are all normal.

Solution

$$\mathcal{A}^* = \mathcal{A} \Longrightarrow \mathcal{A}\mathcal{A}^* = \mathcal{A}^2 = \mathcal{A}^*\mathcal{A},$$
$$\mathcal{A}^* = -\mathcal{A} \Longrightarrow \mathcal{A}\mathcal{A}^* = -\mathcal{A}^2 = \mathcal{A}^*\mathcal{A},$$
$$\mathcal{U} \text{ unitary} \Longrightarrow \langle \mathcal{U}x, \mathcal{U}y\rangle = \langle x, y\rangle \Longrightarrow \mathcal{U}^*\mathcal{U} = I.$$
$$\mathcal{U}^{-1} = \mathcal{U}^*, \qquad \mathcal{U}\mathcal{U}^* = I = \mathcal{U}^*\mathcal{U}.$$

Theorem 4.23 gives normality in all three cases.

Problem 2

Prove Theorem 4.25: $A \in \mathbb{C}^{n\times n}$ is normal if and only if $U^*AU = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ for a unitary U, where the λ_i are the eigenvalues of A.

Solution

By the course definition, normality means having an orthonormal eigenbasis.

$$Au_j = \lambda_j u_j, \quad u_i^* u_j = \delta_{ij} \Longrightarrow U = [u_1 \ \cdots \ u_n], \quad U^*U = I,$$
$$AU = U\Lambda \Longrightarrow U^*AU = \Lambda.$$

$$U^*AU = \Lambda,\ U^*U = I \Longrightarrow AU = U\Lambda \Longrightarrow Au_j = \lambda_j u_j, \quad u_i^* u_j = \delta_{ij}.$$

Problem 3

Prove Theorem 4.27: for $A = (a_{ij}) \in \mathbb{C}^{n\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ counted with multiplicity, $\sum_i |\lambda_i|^2 \le \sum_{i,j} |a_{ij}|^2$, with equality if and only if A is normal.

Solution

By Schur triangularization, choose unitary Q such that $T = Q^*AQ$ is upper triangular and $t_{ii} = \lambda_i$.

$$\sum_{i,j} |a_{ij}|^2 = \operatorname{tr}(A^*A) = \operatorname{tr}(Q^*A^*AQ) = \operatorname{tr}(T^*T)$$
$$= \sum_i |\lambda_i|^2 + \sum_{i<j} |t_{ij}|^2 \ge \sum_i |\lambda_i|^2.$$

$$\text{Equality} \iff t_{ij} = 0\ (i<j) \iff T \text{ is diagonal} \Longrightarrow A \text{ normal.}$$

Conversely, normality permits $T = \Lambda$ by Problem 2, giving equality.

Problem 4

Find a unitary W such that W^*AW is upper triangular for

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{pmatrix}.$$

Solution

$$K = A - I, \quad v_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad u = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad v = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$[v_0\ u\ v]^*[v_0\ u\ v] = I, \qquad Kv_0 = 0, \quad Ku = -\sqrt{3}v, \quad Kv = \sqrt{3}u.$$

$$z_+ = \frac{u + iv}{\sqrt{2}}, \quad z_- = \frac{u - iv}{\sqrt{2}}, \qquad Kz_\pm = \pm i\sqrt{3}z_\pm, \quad z_+^* z_- = 0, \quad \|z_\pm\| = 1.$$

$$\boxed{W = [v_0\ z_+\ z_-] = \begin{pmatrix} 1/\sqrt{3} & (\sqrt{3}+i)/\sqrt{12} & (\sqrt{3}-i)/\sqrt{12} \\ -1/\sqrt{3} & 2i/\sqrt{12} & -2i/\sqrt{12} \\ 1/\sqrt{3} & (-\sqrt{3}+i)/\sqrt{12} & (-\sqrt{3}-i)/\sqrt{12} \end{pmatrix}}.$$

$$W^*W = I, \qquad \boxed{W^*AW = \operatorname{diag}(1, 1 + i\sqrt{3}, 1 - i\sqrt{3})}.$$

Problem 5

Prove Theorem 4.28 directly from the definition: A is normal if and only if $AA^* = A^*A$. Do not quote any other theorem or fact.

Solution

($\Rightarrow$) Take an orthonormal eigenbasis $u_1, \ldots, u_n$ from the definition, with $Au_j = \lambda_j u_j$.

$$u_i^* A^* u_j = (Au_i)^* u_j = \bar{\lambda}_i \delta_{ij} \Longrightarrow A^* u_j = \bar{\lambda}_j u_j.$$

$$AA^* u_j = |\lambda_j|^2 u_j = A^* A u_j \quad (1 \le j \le n) \Longrightarrow AA^* = A^*A.$$

($\Leftarrow$) Induct on the order. The case $n = 1$ is immediate. For $n > 1$, take a root μ of $\det(zI - A)$ and a unit $u \in \ker(A - \mu I)$; put $B = A - \mu I$.

$$BB^* = B^*B, \quad Bu = 0, \qquad \|B^*u\|^2 = u^*BB^*u = u^*B^*Bu = 0 \Longrightarrow A^*u = \bar{\mu}u.$$

$$u^*x = 0 \Longrightarrow u^*Ax = (A^*u)^*x = \mu u^*x = 0, \qquad u^*A^*x = (Au)^*x = \bar{\mu}u^*x = 0.$$

Thus $u^\perp$ is invariant under both A and A^*; the restrictions still commute.

$$V = [u\ Q], \quad V^*V = I, \quad V^*AV = \begin{pmatrix} \mu & 0 \\ 0 & C \end{pmatrix}, \quad V^*A^*V = \begin{pmatrix} \bar{\mu} & 0 \\ 0 & C^* \end{pmatrix} \Longrightarrow CC^* = C^*C.$$

Induction gives an orthonormal eigenbasis $u_2, \ldots, u_n$ on $u^\perp$. Together with u, it is an orthonormal eigenbasis of A, so A is normal.

Problem 6

Prove Theorem 4.29: for $A \in \mathbb{C}_n^{n\times n}$ (that is, $\operatorname{rank} A = n$), there is a unique factorization $A = UR$, where U is unitary and R is upper triangular with positive diagonal entries.

Solution

Existence.

$$A = [a_1 \ \cdots \ a_n], \qquad v_j = a_j - \sum_{i=1}^{j-1} u_i(u_i^* a_j), \qquad r_{jj} = \|v_j\| > 0, \quad u_j = \frac{v_j}{r_{jj}}.$$

$$r_{ij} = u_i^* a_j \ (i<j), \quad r_{ij} = 0 \ (i>j), \qquad u_i^* v_j = u_i^* a_j - \sum_{k<j} \delta_{ik} u_k^* a_j = 0 \ (i<j).$$

Independence of the columns ensures $v_j \neq 0$. Hence

$$u_i^* u_j = \delta_{ij}, \qquad a_j = \sum_{i=1}^{j} u_i r_{ij} \quad \Longrightarrow \quad \boxed{A = UR, \quad U^*U = I, \quad r_{jj} > 0}.$$

Uniqueness.

$$A = U_1R_1 = U_2R_2 \quad \Longrightarrow \quad M = U_2^*U_1 = R_2R_1^{-1}, \qquad M^*M = I.$$

M is upper triangular, with $m_{jj} = (R_2)_{jj}/(R_1)_{jj} > 0$. Its first column and column orthogonality give

$$|m_{11}|^2 = 1 \Longrightarrow m_{11} = 1, \qquad \bar{m}_{11}m_{1j} = 0 \Longrightarrow m_{1j} = 0 \ (j>1).$$

Repeat on the trailing principal block:

$$M = I \quad \Longrightarrow \quad U_1 = U_2, \qquad R_1 = R_2.$$

Problem 7

Prove Theorem 4.30(1): a normal matrix A is Hermitian if and only if all its eigenvalues are real.

Solution

$$A = U\Lambda U^*, \qquad A^* = U\bar{\Lambda}U^*.$$

$$\boxed{A = A^* \iff \Lambda = \bar{\Lambda} \iff \lambda_j = \bar{\lambda}_j \ (\forall j) \iff \lambda_j \in \mathbb{R} \ (\forall j)}.$$

Problem 8

Prove Theorem 4.31(1): $A \in \mathbb{C}^{n\times n}$ is Hermitian if and only if $x^*Ax \in \mathbb{R}$ for every $x \in \mathbb{C}^n$.

Solution

($\Rightarrow$) $\overline{x^*Ax} = x^*A^*x = x^*Ax \Longrightarrow x^*Ax \in \mathbb{R}$.

($\Leftarrow$) $H = A - A^*$: $x^*Hx = x^*Ax - \overline{x^*Ax} = 0$.

$$x = e_i \Longrightarrow h_{ii} = 0,$$
$$x = e_i + e_j \Longrightarrow h_{ij} + h_{ji} = 0,$$
$$x = e_i + ie_j \Longrightarrow ih_{ij} - ih_{ji} = 0.$$

$$h_{ij} = h_{ji} = 0 \quad (\forall i,j) \quad \Longrightarrow \quad H = 0 \quad \Longrightarrow \quad \boxed{A = A^*}.$$

Problem 9

For Hermitian A and B, prove that AB is Hermitian if and only if $AB = BA$ (Property 5 on textbook page 168).

Solution

$$(AB)^* = B^*A^* = BA \quad \Longrightarrow \quad \boxed{AB \text{ is Hermitian} \iff AB = (AB)^* \iff AB = BA}.$$